

Running the `coverage run -m unittest tests/unit_tests.py`, followed by `coverage report` the first time:

```
● $ coverage report
```

Name	Stmts	Miss	Cover
-----	-----	-----	-----
app.py	159	52	67%
pymongo_get_db.py	14	2	86%
tests\unit_tests.py	75	2	97%
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TOTAL	248	56	77%

If not otherwise specified, the untested functions, return cases and redirects were not accounted for in our tests since in assignments 1, 2 and 3, we primarily focused on testing the actions that the user can take that affect the database, with less focus on functions that make the website work properly. In other words, we tested for the expected result for a defined user action rather than the error case, because it is unlikely that an error would occur when we have implemented measures to prevent them. An HTTP 400 code won't break the application, however if the data manipulation functions weren't working as intended, this would be catastrophic for the application.

For example, we tested whether drugs could be added to the database, but we didn't test if an error was returned if no drug was provided, since this isn't what the user would want to be doing primarily, and the form allowing users to submit data already requires that all the entries be filled. This isn't to say that these tests are unimportant, but rather that we had bigger priorities. Additionally, the test cases that were written in assignment 1 were primarily focused on the intended purpose of the website.

Misses in the `pymongo_get_db.py` file:

- The exception for a failed db connection in the try except statement was missed in the tests

Misses in the `app.py` file:

- index function
- dashboard function
- add_sample_drug function
 - This was not tested as it wasn't used in the website
- drug_table function
- browse_drug function
- drug_edit function
 - return 404 case for "Drug not found"
 - return 400 case for "Invalid stock value"
 - redirect to /drug-info case
 - GET method case
- new_drug function

- return 400 case for "Invalid stock value"
- redirect to /drug-info case
- GET method case
- delete_drug function
 - return 404 case for "Drug not found"
 - GET method case
- drug-info function
- add_sample_order function
 - This was not tested as it wasn't used in the website
- new_order function
 - GET method case
- edit_order function
 - return 404 case for "Order not found"
 - return 400 case for "Invalid date value"
 - GET method case
- delete_order function
 - return 404 case for "Order not found"
 - GET method case
- main method
 - Didn't test if the app runs on port 5001

Misses in the tests/unit_tests.py file:

- main method
 - Didn't test if the test file outputs "All tests have run." upon completion of tests

Notice how most of the misses above are either for functions that aren't relating to the primary use cases of the website. For the functions that are relating to these use cases, the untested parts are exclusively missing error cases or GET method cases, which are also not part of the primary use cases.

After writing more tests to cover most of the above misses, the new result when running the coverage tool again is:

Name	Stmts	Miss	Cover
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app.py	159	15	91%
pymongo_get_db.py	14	0	100%
unit_tests.py	190	3	98%
-----	-----	-----	-----
TOTAL	363	18	95%

With the new tests, the only untested parts of app.py are either GET method cases or return 400 cases.